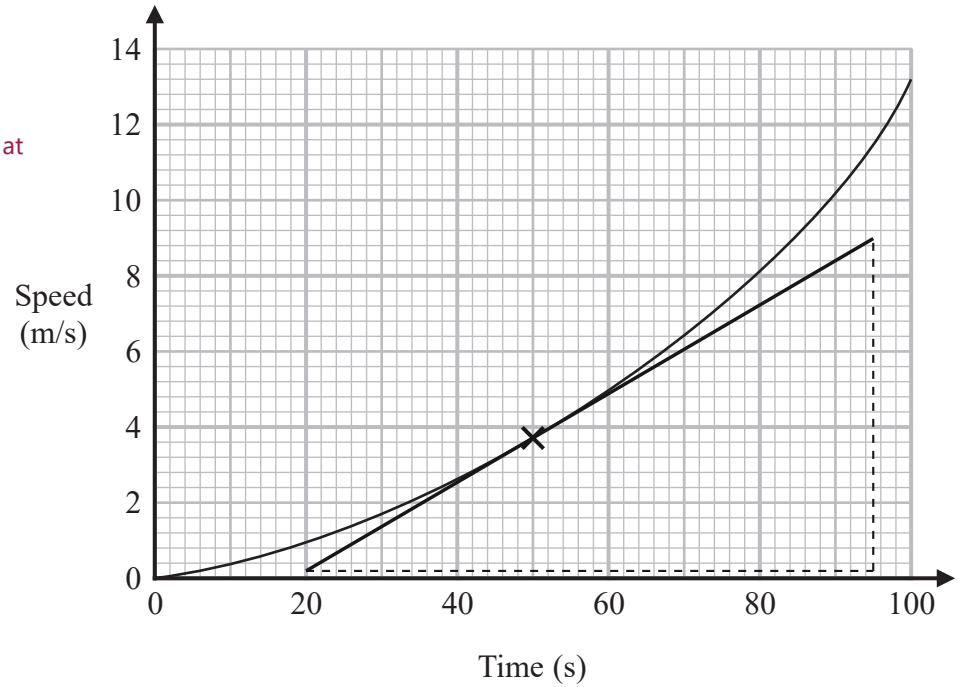
**15** Here is a speed-time graph.




**1 mark**
Draw tangent at t = 50.

(a) Calculate an estimate of the gradient of the graph at time 50 seconds.
You must show how you get your answer.

Tangent points: (20, 0.19) and (95, 8.99)

Gradient = rise ÷ run

**1 mark**
Change in speed ÷ change in time.

$$\frac{8.99 - 0.19}{95 - 20} \approx 0.117 \text{ m/s}^2$$

**1 mark**
About 0.12 m/s².

**(3)**

(b) Describe what your answer to part (a) represents.

The acceleration at 50 seconds.

**1 mark**
Rate of change of speed.

**(1)**

**(Total for Question 15 is 4 marks)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN TH